

Regulators of ECM Structure Enable Functional Adaptation to Tensile Loading in Tendon Explants

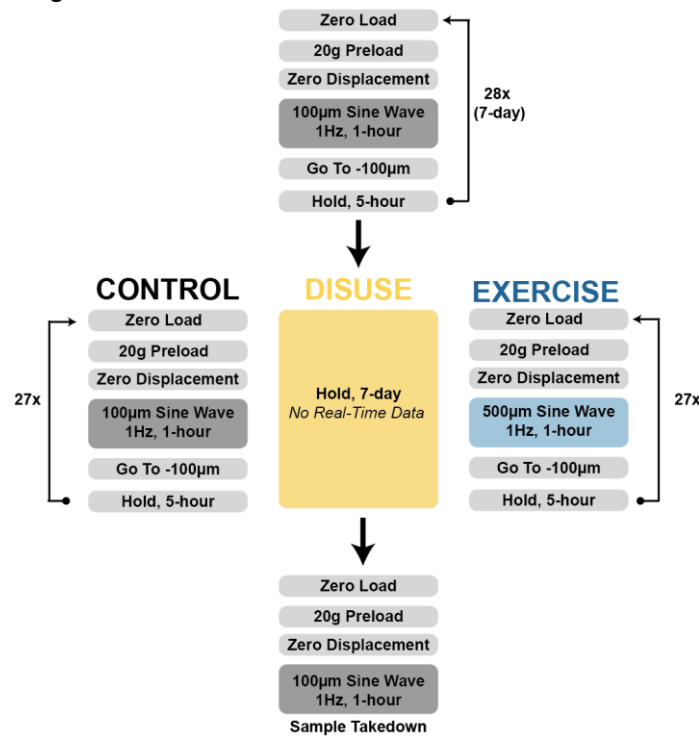
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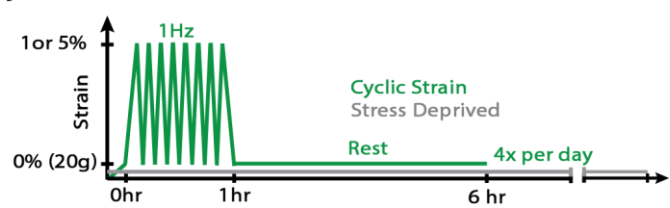
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Supporting Information

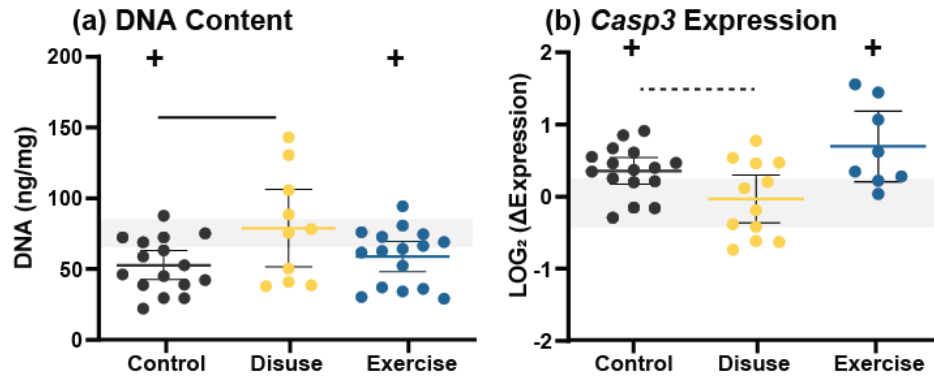
(a) Loading Flow Chart



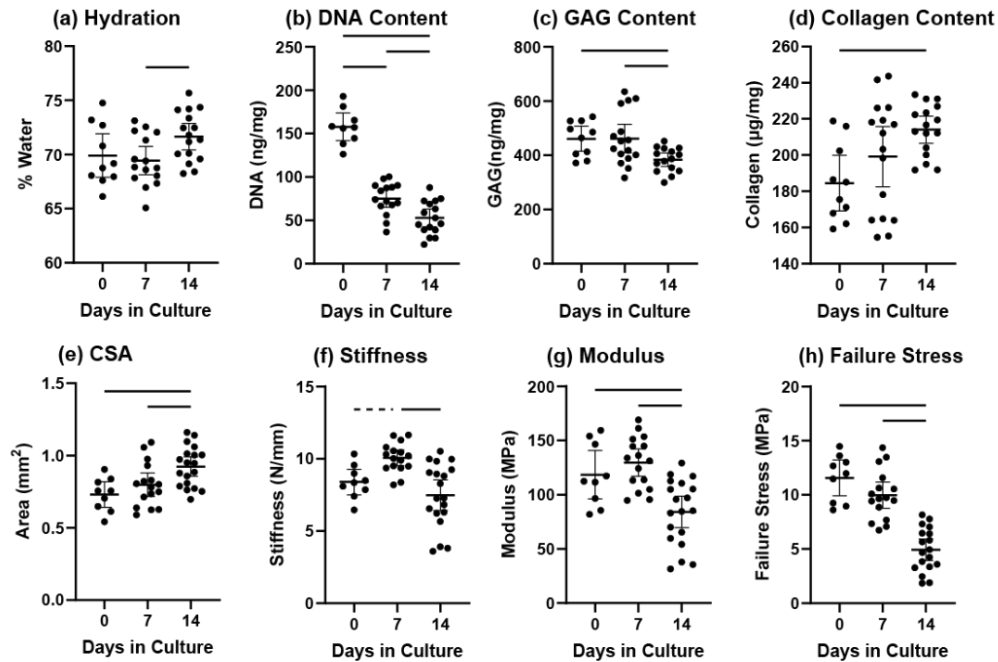
(b) Cyclic Strain Waveform



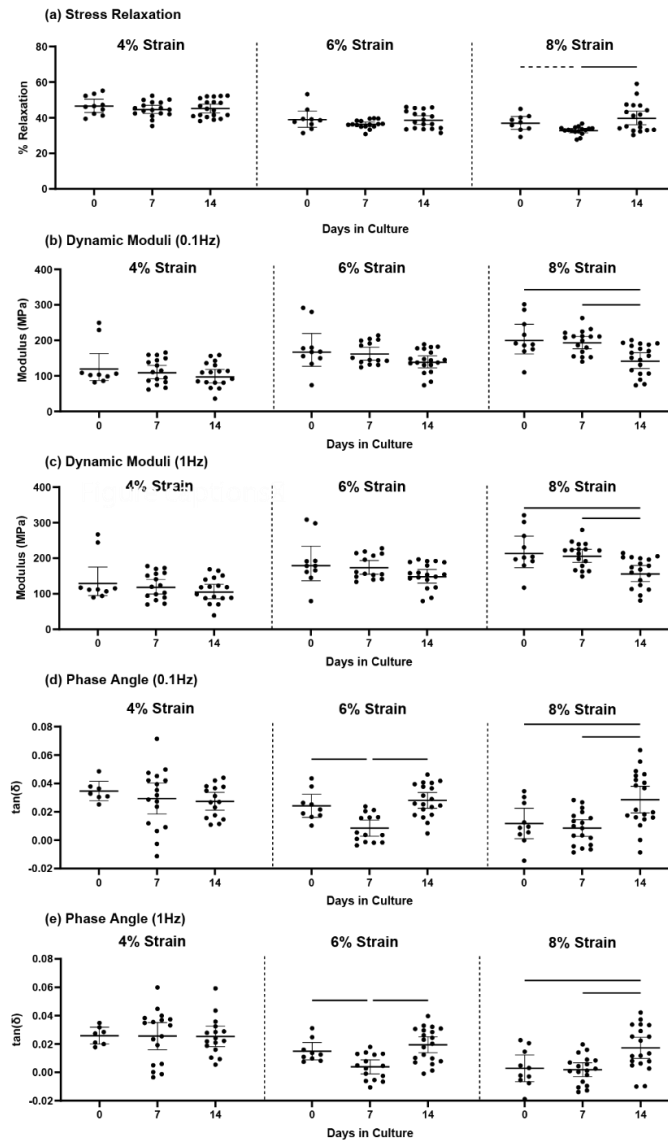
Supporting Figure S1. (a) Bioreactor loading flow chart and (b) cyclic waveform schematic.



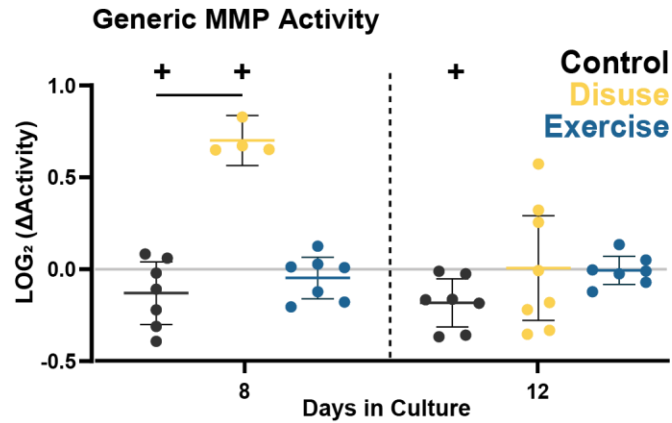
Supporting Figure S2. (a) DNA content and (b) *Casp3* gene expression of day 14 show no significant injury response for control, disuse, or exercise tendons. Gene expression data presented as relative expression to day 7 pre-intervention level. Data are shown as individual points with mean $\pm$ 95% confidence interval. Significant ($p < 0.05$) comparisons versus control are indicated with a solid line while trends ($p < 0.1$) are indicated with a dashed line. Grey horizontal bands show the day 7 pre-intervention reference levels ($\pm$ 95% CI), and (+) symbols indicate significant differences from day 7 ($p < 0.05$).



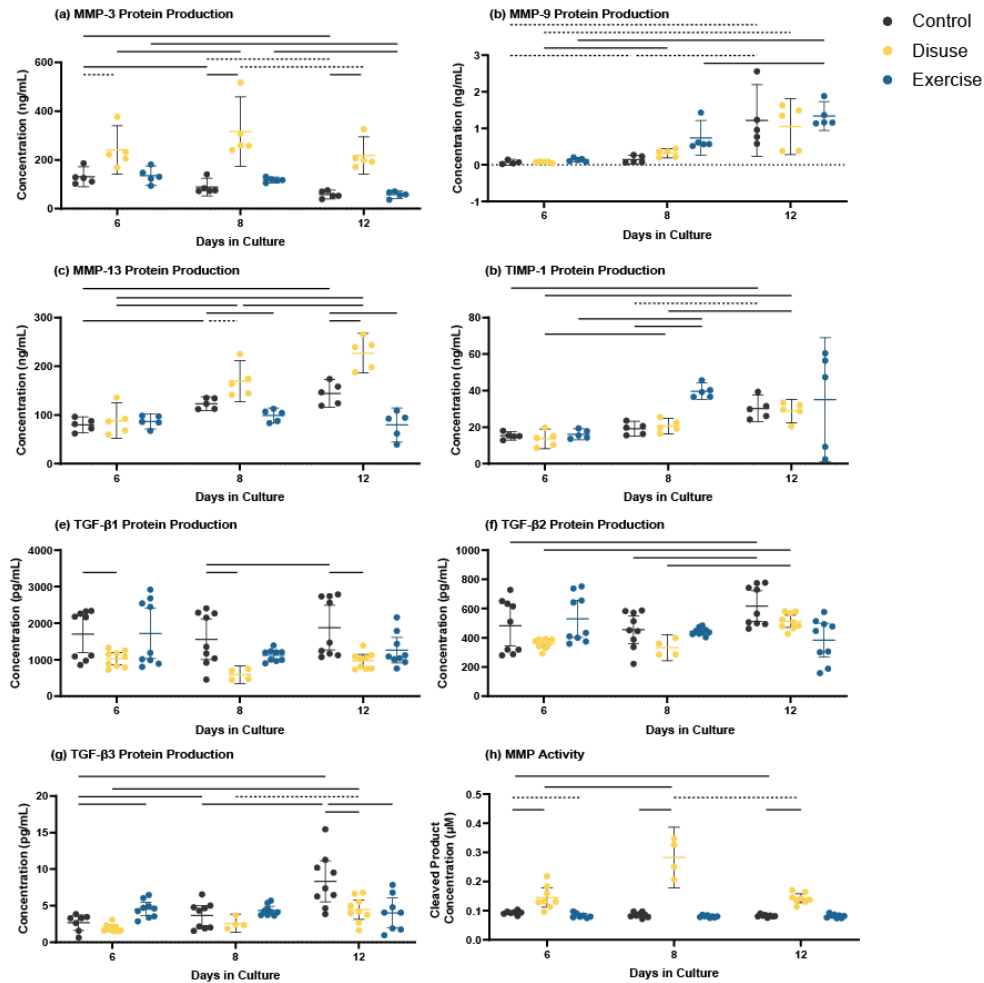
Supporting Figure S3. (a) Hydration, (b) DNA content, (c) GAG content, (d) collagen content, (e) CSA, (f) stiffness, (g) elastic modulus, and (h) failure stress of tendons at day 0 (native), day 7, and day 14 of *ex vivo* control loading (1% cyclic strain). Data are shown as individual points with mean $\pm$ 95% confidence interval. Significant ($p < 0.05$) comparisons are indicated with a solid line while trends ($p < 0.1$) are indicated with a dashed line.



Supporting Figure S4. (a) Stress relaxation, (b,c) dynamic modulus, and (d,e) phase angle of tendons at day 0 (native), day 7, and day 14 of *ex vivo* control loading (1% cyclic strain). Dynamic testing was conducted at 4%, 6%, and 8% grip strain with 0.1 Hz and 1Hz frequency sweeps. Data are shown as individual points with mean $\pm$ 95% confidence interval. Significant ($p < 0.05$) comparisons are indicated with a solid line while trends ($p < 0.1$) are indicated with a dashed line.



Supporting Figure S5. Fold change in generic MMP activity in spent culture medium at days 8 and 12 relative to day 6 pre-intervention levels. Activity measures are paired values collected longitudinally over the culture period for each experiment. Data are shown as individual points with mean $\pm$ 95% confidence interval. Significant ($p < 0.05$) comparisons versus control are indicated with a solid line. (+) symbols indicate significant differences from day 6 ($p < 0.05$). Raw data in Supporting Figure S6.



Supporting Figure S6. Concentrations of (a) MMP-3, (b) MMP-9, (c) MMP-13, (d) TIMP-1, (e) TGF- β 1, (f) TGF- β 2, and (g) TGF- β 3 in spent culture medium at days 6, 8, and 12. (h) Concentration of cleaved MMP product representing MMP activity in culture medium. Concentrations are paired values collected longitudinally over the culture period for each experiment. Data are shown as individual points with mean $\pm$ 95% confidence interval. Multiple comparisons are performed with respect to both time and intervention. Significant ($p < 0.05$) pairwise comparisons are indicated with a solid line while trends ($p < 0.1$) are indicated with a dashed line.

Supporting Table S1. Murine gene names and primer forward/reverse sequences used for quantitative gene expression analysis.

Target	Gene Name	Forward (5'-> 3')	Reverse (5'-> 3')
Beta Actin	<i>Actb</i> ¹	GGCTGTATTCCCCTCCATCG	CCAGTTGGTAACAATGCCATGT
Caspase-3	<i>Casp3</i> ²	ATGGGAGCAAGTCAGTGGAC	CGTACCAGAGCGAGATGACA
IL-6	<i>Il6</i> ³	TAGCTACCTGGAGTACATGAAGAACA	TGGTCCTTAGCCACTCCTTCTG
TNF- α	<i>Tnfa</i> ⁴	GTCTCAGCCTCTTCTCATTC	CATAGAACTGATGAGAGGGA
Collagen 1	<i>Col1a1</i> ³	GACATGTTTCAGCTTTGTGGACCTC	GGGACCCCTTAGGCCATTGTGTA
Collagen 3	<i>Col3a1</i> ⁵	CAGAGCAACGGTCATACTCATTACACC	CAGCAACAGCAGAAGAGAAGCACC
Biglycan	<i>Bgn</i> ⁵	TTTCTGAGCTTCGCAAGGATG	GGGCGTAGAGGTGCTGGAG
Decorin	<i>Dcn</i> ³	CTATGTGCCCTACCGATGC	CAGAACTGCACCACTCGAAG
Fibromodulin	<i>Fmod</i> ³	CTCCAACCCAAGGAGACCAG	GGATCCACCAGTGAGAGTCTTC
TGF- β 1	<i>Tgfb1</i> ⁶	ATTCTGGCGTTACCTTGG	CCTGTATTCCGTCTCCTTGG
MMP-1	<i>Mmp1</i> ⁷	TCAACCAGGCCAAGGTATTG	ATGAGCAGCCACGAGAAATAG
MMP-3	<i>Mmp3</i> ²	ACATGGAGACTTTGTCCCTTTTG	TTGGCTGAGTGGTAGAGTCCC
MMP-13	<i>Mmp13</i> ²	TCAGTCTCTTCACCTCTTTTGGGAATCC	TCAGTTTCTTTATGGTCCAGGCGATG

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Supporting Table S2: Details of variables included in the correlation analysis.

Variable	Full Name	Units	Assay	Timepoint
E	Elastic Modulus	Mpa	Mechanical Testing	Day 14
σ -max	Maximum Stress	MPa	Mechanical Testing	Day 14
ϵ -max	Maximum Strain	%	Mechanical Testing	Day 14
E*	Dynamic Modulus	MPa	Mechanical Testing	Day 14
SR	Stress Relaxation	%	Mechanical Testing	Day 14
Tan δ	Tangent of Phase Angle	-	Mechanical Testing	Day 14
AI	Collagen Alignment Index	[0,1]	SHG Imaging	Day 14
³ H	3H-Proline Incorporation	pmol/hr/mg	Biosynthesis/Composition	Day 14
COL	Collagen Content	μ g/mg	Biosynthesis/Composition	Day 14
³⁵ S	35S-Sulfate Incorporation	pmol/hr/mg	Biosynthesis/Composition	Day 14
GAG	Glycosaminoglycan Content	ng/mg	Biosynthesis/Composition	Day 14
H ₂ O	Water Content	%	Biosynthesis/Composition	Day 14
[TGF- β 1]	TGF- β 1 Concentration in Media	pg/mL	Multiplex Immunoassay	Day 12
[TGF- β 2]	TGF- β 2 Concentration in Media	pg/mL	Multiplex Immunoassay	Day 12
[TGF- β 3]	TGF- β 3 Concentration in Media	pg/mL	Multiplex Immunoassay	Day 12
[MMP-3]	MMP-3 Concentration in Media	ng/mL	Multiplex Immunoassay	Day 12
[MMP-9]	MMP-9 Concentration in Media	ng/mL	Multiplex Immunoassay	Day 12
[MMP-13]	MMP-13 Concentration in Media	ng/mL	Multiplex Immunoassay	Day 12
[TIMP-1]	TIMP-1 Concentration in Media	ng/mL	Immunoassay	Day 12
MMPa	Generic MMP Activity in Media	Cleavage Product μ M	Activity Assay	Day 12
<i>Col1</i>	<i>Col1</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Col3</i>	<i>Col3</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Bgn</i>	<i>Bgn</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Fmod</i>	<i>Fmod</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Dcn</i>	<i>Dcn</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Mmp1</i>	<i>Mmp1</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Mmp3</i>	<i>Mmp3</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Mmp13</i>	<i>Mmp13</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Tgfb1</i>	<i>Tgfb1</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Il6</i>	<i>Il6</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Tnfa1</i>	<i>Tnfa1</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14
<i>Casp3</i>	<i>Casp3</i> Gene Expression	Relative Expression (<i>Actb</i>)	qPCR	Day 14

Supporting Table S3. Pearson r correlation matrix.

[illegible]

Supporting Table S4. Correlation analysis p-values.

[illegible]

Supporting Table S5. Correlation analysis sample sizes (n).

[illegible]